



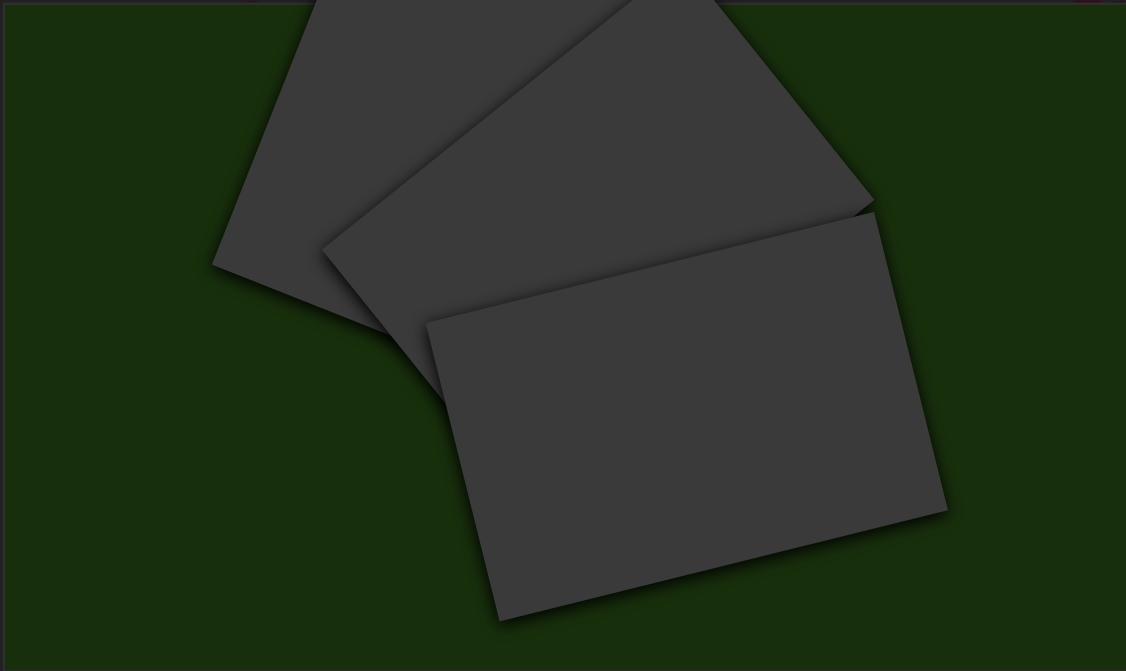
Project Proposal

Jack, Christian J, Christopher B, Julia

Overall Goal

Create a video game with an AI rival that learns to exploit the player's weaknesses.

We've chosen to make a deck-building roguelike inspired by slay the spire.



Features + Requirements

FEATURES

- System for gamified card-based duels
- Procedurally-generated card library
- AI which learns to construct its own deck by adapting to the player's deck and play style
- Scaling increase in the strength of generated cards over time

REQUIREMENTS

- Image and audio assets
- Asset loading
- HTML5/JS game engine
- Card game system
- Procedural card library system
- Custom AI/ML system

Scope Boundary

IN SCOPE

- Basic card game system
- Procedurally generated card library
- Traditional machine learning algorithms
- Prototype audio/art curation/production

NOT IN SCOPE

- Generative AI, natural language AI
- Multiplayer, networking, leaderboard
- Fully polished art, voice acting, story, dialog

Task Breakdown

1. Project setup
 - a. Install tools, create repository, JIRA for issue tracking, etc.
2. Gather prototype assets
 - a. Placeholder images, names, sounds, cards, etc. Minimal, for testing only.
3. Game systems programming
 - a. Reuse well-known game programming patterns to write an engine fit for our game
4. ML integration
 - a. Test multiple methods, evaluate results
5. Content/asset expansion
 - a. Expand art/assets as well as game mechanics
 - b. Improve integration of AI with these features

Team Responsibilities

- Jack
 - Engine programming/tooling lead, audio lead, ML architect
- Christian J
 - Debug, playtest, gameplay feature development, AI integration
- Christopher B
 - Procedural generation, gameplay balancing, AI balancing
- Julia
 - Creative direction, art lead, UX/UI lead